Project 1: NOvA Atmospheric Neutrino Analysis

- Big Physics Question: Do neutrinos experience forces beyond the Standard Model when traversing matter?

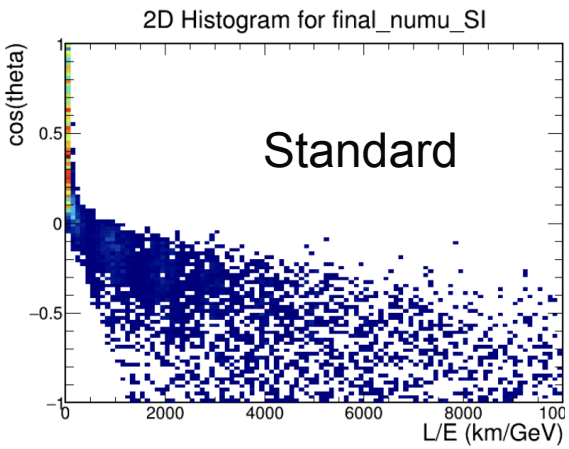
- Can NOvA see atmospheric neutrinos through the cosmic ray noise? (NOvA's pioneering attempt)

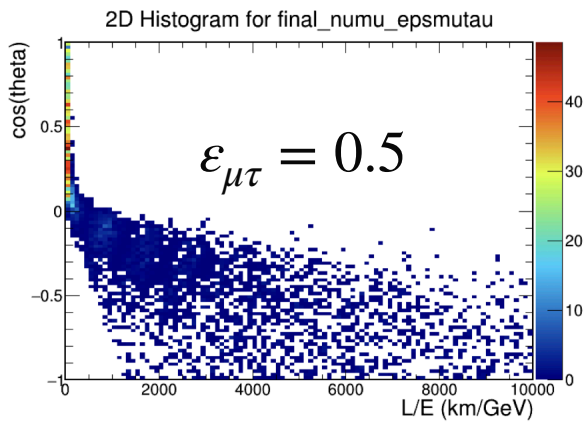
Why Atmospheric Neutrino Matters for NOvA

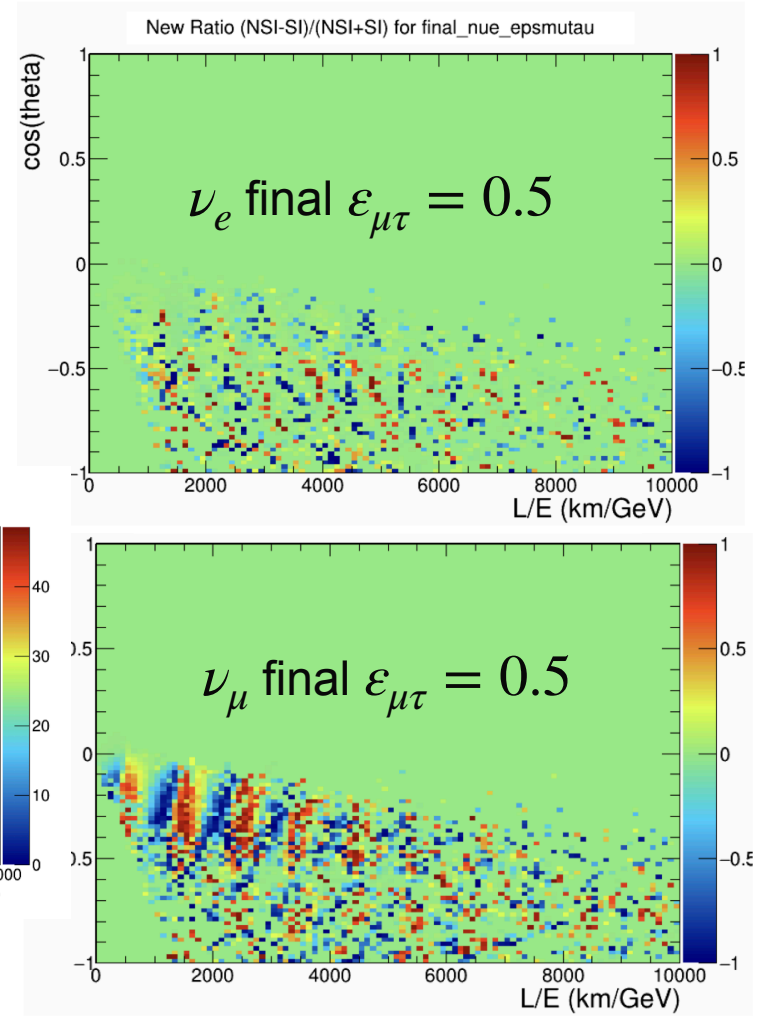
- NOvA has been collecting atmospheric data for nearly a decade, yet no analysis has been completed.
- This created a real physics opportunity: no one had quantified whether atmospheric neutrinos at a surface detector like NOvA were actually usable.
- My goal was to answer this open question: Can NOvA realistically extract atmospheric neutrino signals with useful purity and resolution?
- My work provides the first complete attempt to answer this.

NSI Signals in NOvA Monte Carlo

- Apply oscillogram to NOvA Monte Carlo simulation
- NSI modifies both final state distributions
- NOvA FD MC can identify NSI signal



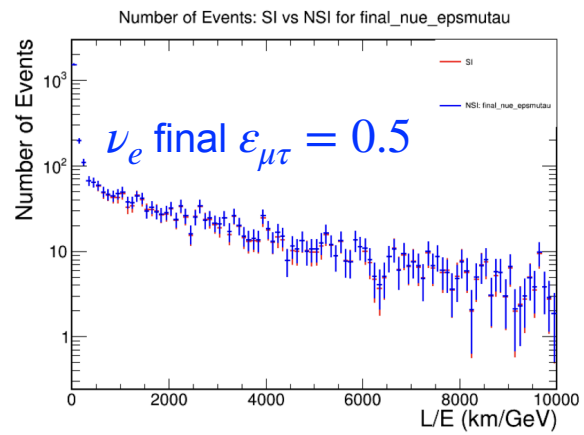


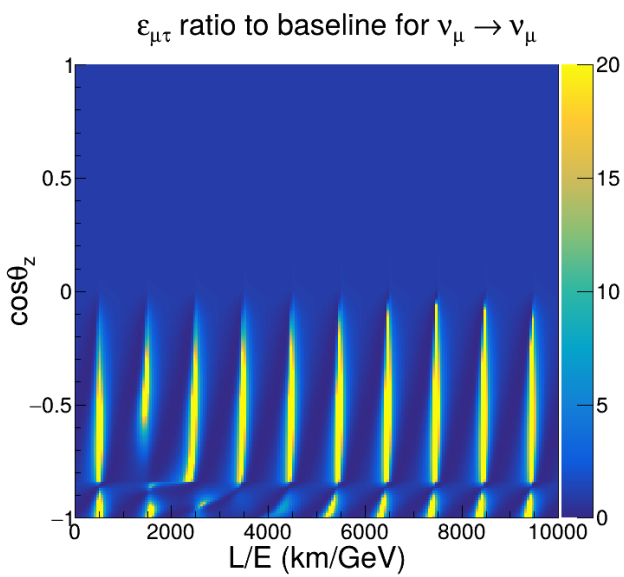


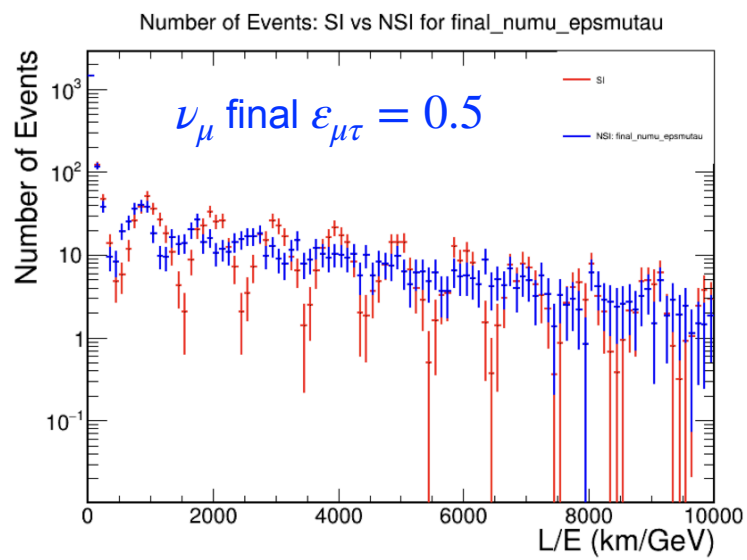
(b) With NSI: $\varepsilon_{\mu\tau} = 0.5$

NSI Signals in 1D Projection

- Event rate match Oscillogram pattern
- NSI show significant deviation from SI after statistical error



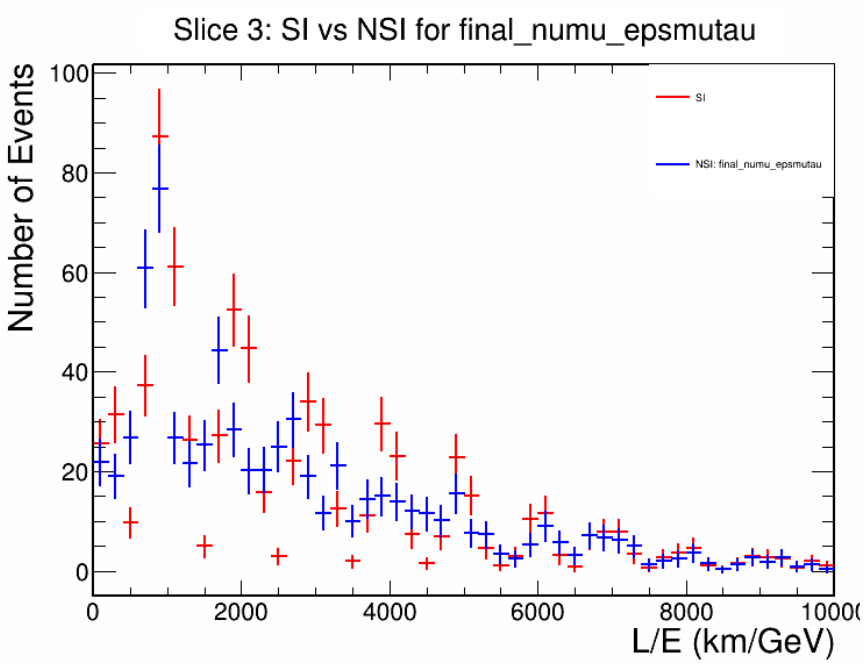


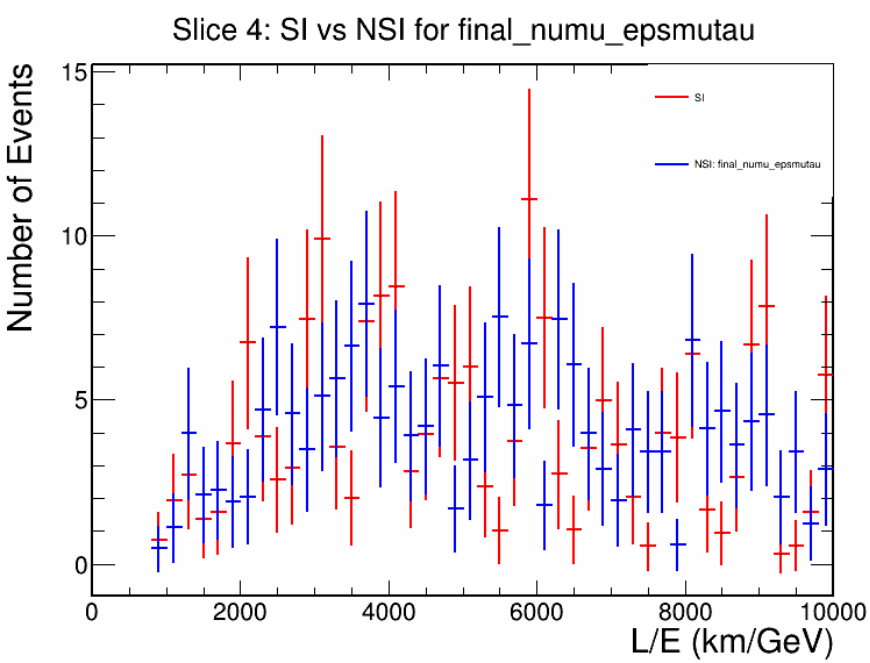


NSI Signals in 1D Slices

- NSI effect is angle dependent
- Slice $0 < cos(\theta) < 1$ show no NSI impact
- Challenge: Can we extract this subtle signal from cosmic ray background?

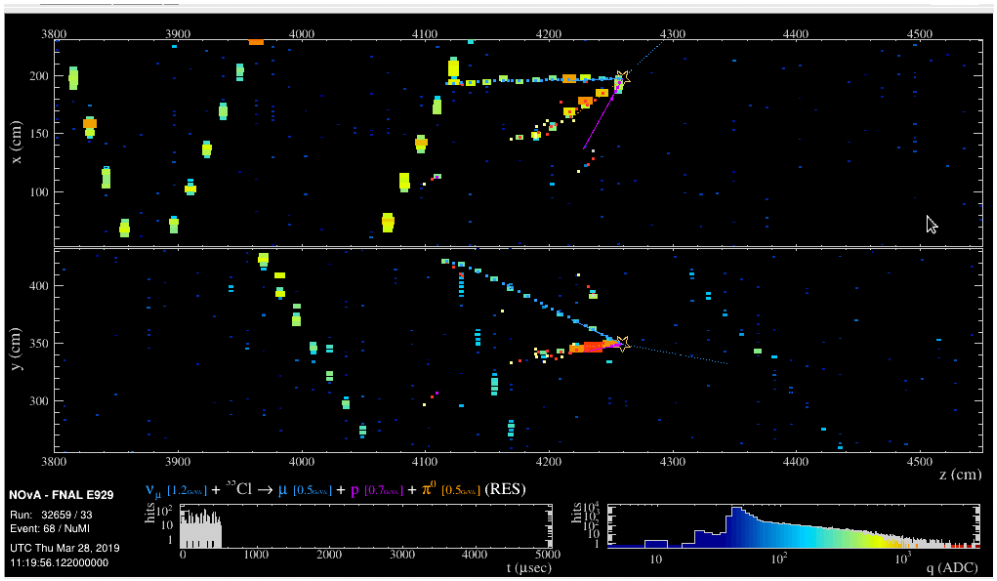
Slice 3: $-0.5 < cos(\theta) < 0$ (horizontal to upward)
Slice 4: $-1 < cos(\theta) < -0.5$ (strongly upward) - maximum Earth matter traversal





Reconstruction

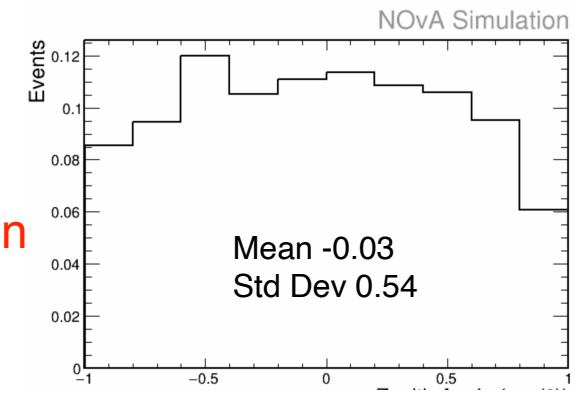
- Goal: extract neutrino energy, direction and particle ID from detector hits
- Reconstruction chain:
  - Slicer: cluster hits by space-time proximity
  - Vertex finding
  - Prong identification
  - 3D trajectory reconstruction
- Particle identification
  - Muon track identify
  - CNN: flavor identification
- Zenith angle reconstruction: critical for L/E and NSI sensitivity

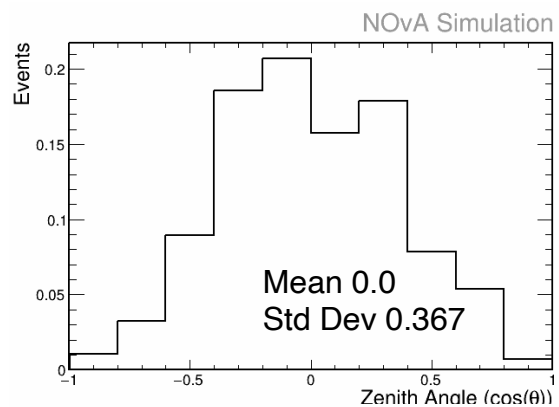


Angle Reconstruction

- Energy weighted vector sum of reconstructed prongs: $\vec{P}_{\text{tot}} = \dfrac{\sum_i E_i \vec{\mathbf{d}}_i}{\left|\sum_i E_i \vec{\mathbf{d}}_i\right|}$
- Zenith angle: $cos\theta \ = \vec{P}_{tot} \cdot \hat{y}$ (vertical direction)
- Poor ν_μ angular residual

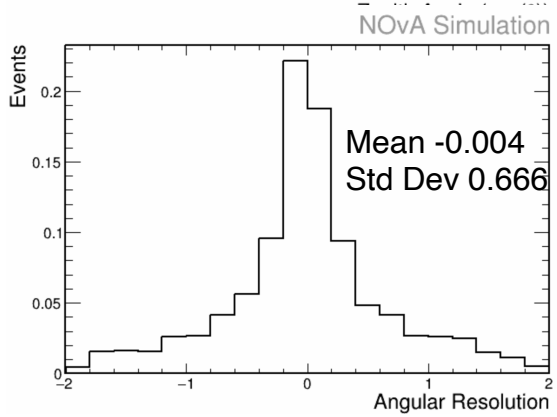
ν_μ θ_z distribution

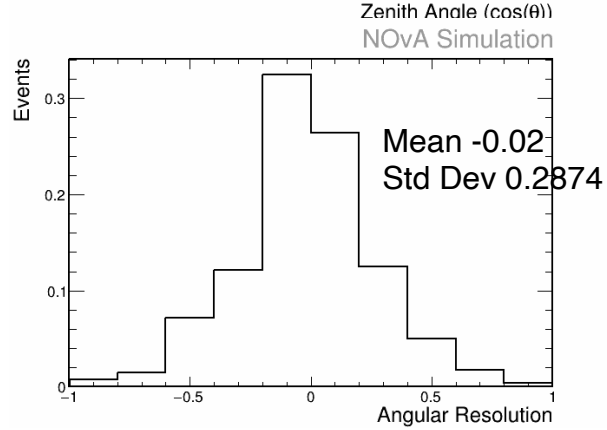




ν_e distribution

ν_μ residual

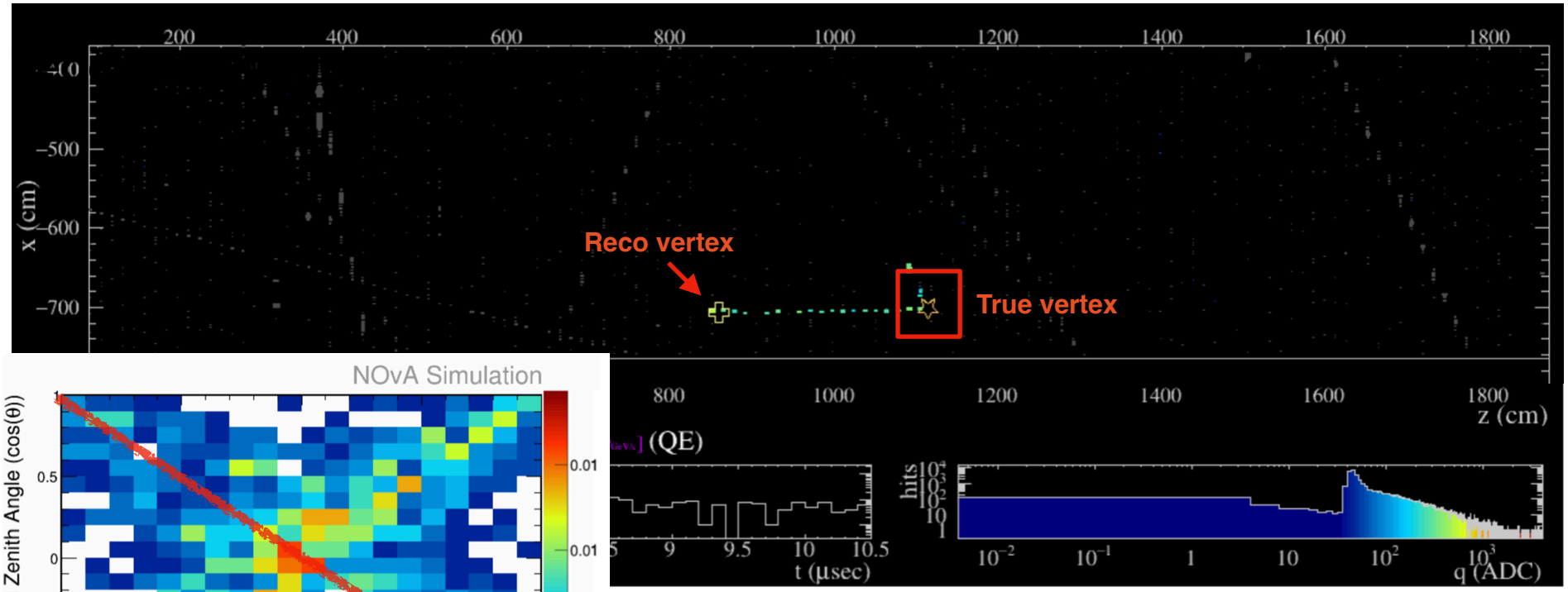


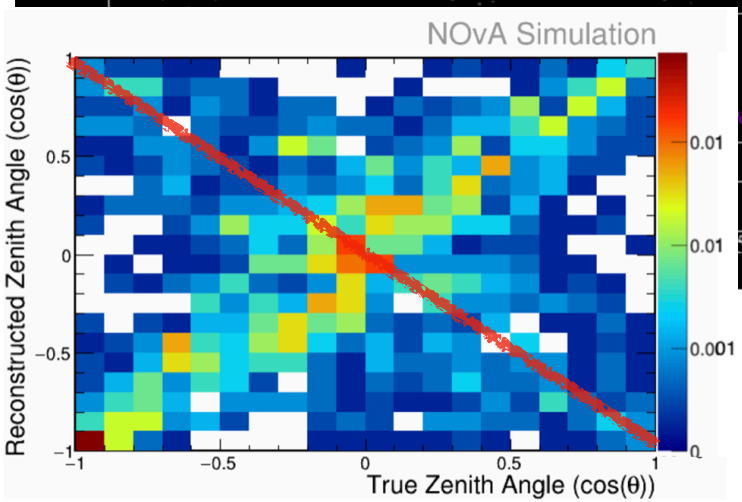


ν_e residual

Reconstructed neutrino direction vs true neutrino direction

- Vertex reconstruction is biased toward upstream of beam direction





Optimistic ν_μ Angle Resolution

- Assuming we can overcome beam direction bias, what would resolution be?
- We study this by selecting well reconstructed events, where the reconstructed vertex is within 30cm of the true vertex
- Ways to achieve this in data: train vertex algorithm on simulated atmospheric neutrinos; look for hadronic activity to identify real vertex

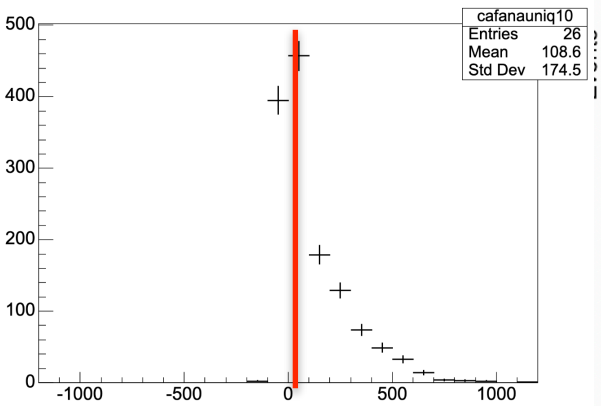
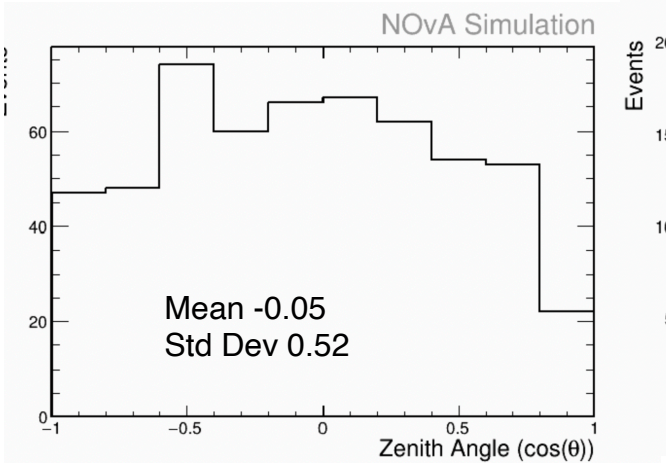
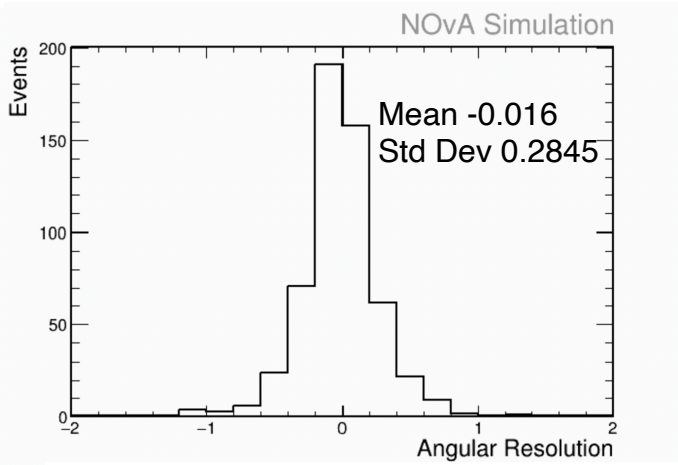


Figure 5.15: true neutrino vertex z position - reco neutrino vertex z position





ν_μ anglar distribution

ν_μ angular residual

Selection Efficiency Summary

- Efficiency = $\frac{\nu_{(reco\ with\ cut)}}{\nu_{(pass\ NNBar)}}$
- Out of all neutrinos triggered, what fraction made it through the full analysis chain

Parameter	ν_e CC	ν_μ CC
Events after NNBar trigger	2,748	2,642
Events selected	200	862
Selection efficiency	**7.3%**	**32.6%**
NC contamination	1.9%	3.7%
Flavor cross-contamination	0.7%	4.0%

Cut flow

- Systematic selection optimization for each flavor
- ν_μ selection (top)
  - Quality cut: maintain high efficiency (93%)
  - ReMId: 35x NC rejection at cost of efficiency (42%)
  - Containment: critical for cosmic 128x rejection
- ν_e selection (bottom)
  - Energy/topology core cuts: 32x cosmic rejection
  - PID: extremely powerful 5070x cosmic, 157x NC rejection
  - Only 7.3% efficiency but excellent bkg rejection

Table 5.2: Cut flow showing signal efficiency and background rejection for ν_μ CC selection

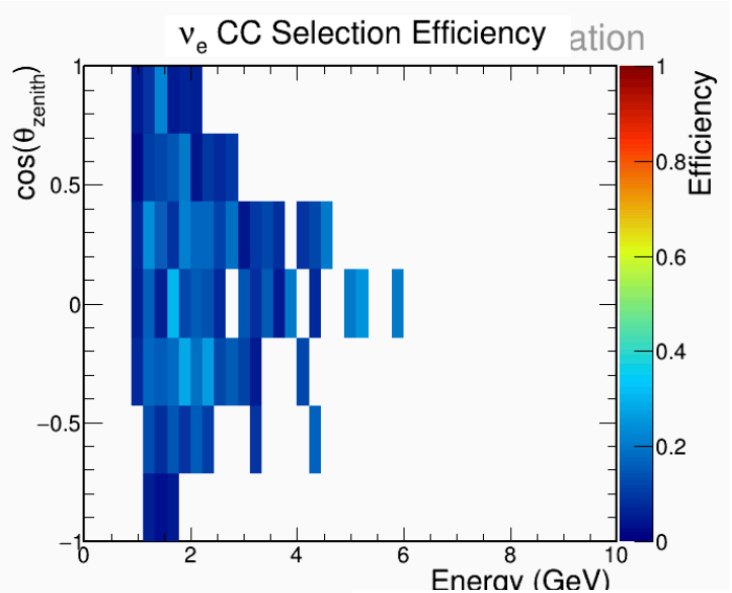
Cut Stage	Signal	Eff.	NC Bkg	Rej.	Cosmic Bkg	Rej.
After NNBar trigger	2662	—	1252	—	8.48×10^8	—
+ kHas1Prongs	2612	98.1%	1229	1.0×	8.48×10^8	1.0×
+ kNumuBasicQuality	2486	95.2%	1130	1.1×	7.79×10^8	1.1×
+ kNumuNCRej (RemID>0.75)	1042	41.9%	32	35.3×	7.79×10^8	1.0×
+ Containment	862	82.7%	32	1.0×	4.53×10^6	172.1×

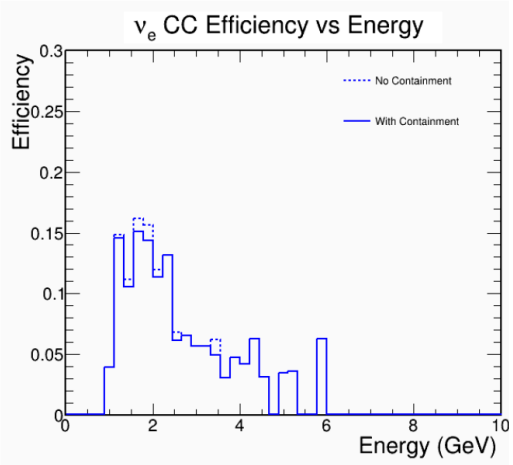
Table 5.3: Cut flow showing signal efficiency and background rejection for ν_e CC selection

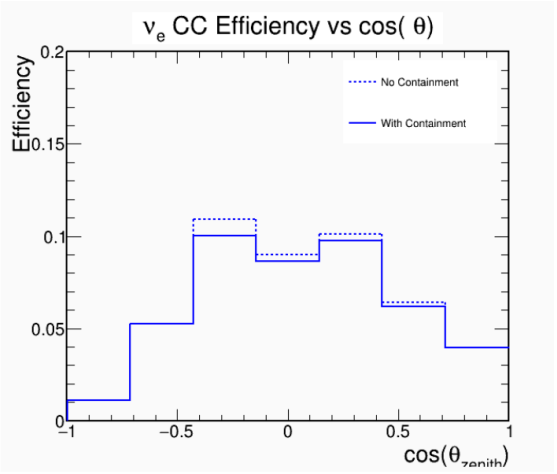
Cut Stage	Signal	Eff.	NC Bkg	Rej.	Cosmic Bkg	Rej.
After NNBar trigger	2763	—	1252	—	1.11×10^9	—
+ kHas1Prongs	2713	98.2%	1229	1.0×	1.11×10^9	1.0×
+ kNue2024FDBasicQuality	2386	87.9%	1013	1.2×	1.02×10^9	1.1×
+ kModifiedNue2024CorePart	1479	62.0%	627	1.6×	3.22×10^7	31.8×
+ kNue2024PID	209	14.1%	4	156.8×	6.35×10^3	5069.6×
+ Containment	200	95.70%	3	1.4×	6.35×10^3	1.0×

$\nu_e CC$ Selection Efficiency Continued

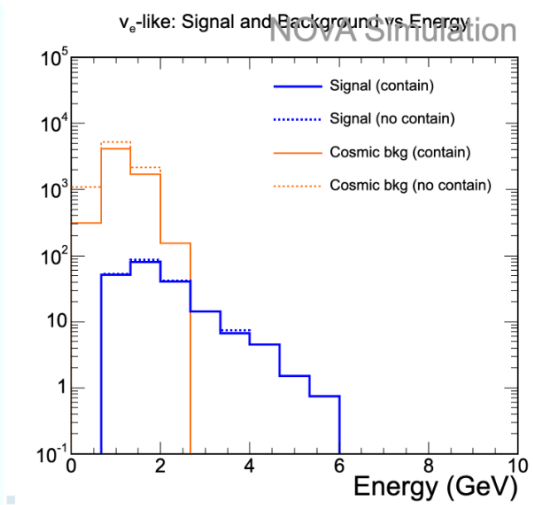
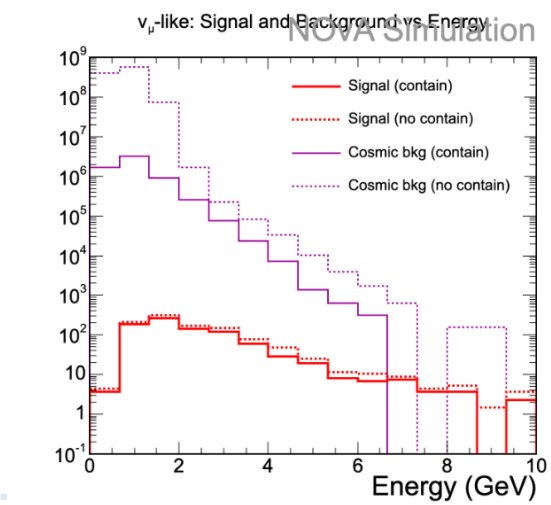
- 200 selected $\nu_e CC$
- 2748 total $\nu_e CC$
- Peak efficiency: ~15% at 1-2 GeV
- 7.3% total selection efficiency
- Strong zenith angle dependence







Purity

- Reality:
  - ν_e: 1 in 33 events is signal
  - ν_μ: 2 in 7000 events is signal
- Even with aggressive cuts, cosmic overwhelm ν_μ signals
- $E_{\nu_e} > 2.3$ GeV have good purity, useable over 4GeV
- $E_{\nu_\mu} > 6.2$GeV good purity, but hard to contain

Component	ν_e-like	ν_μ-like
Signal Events (8 years)		
Atmospheric ν signal	200	862
Cosmic Ray Background		
No containment cut	9×10^3	1×10^9
With containment cut	6×10^3	6×10^6
Rejection factor	1.4×	172×
Purity		
Purity (no containment)	2.36%	0.0001%
Purity (with containment)	3.06%	0.0141%